

Charles O'Byrne

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Eagle Scout

EDUCATION

University of Iowa

Iowa City, IA

Graduation: May 2027

Bachelor of Science in Engineering

Expected Minors: Math, Computer Science,

Electrical Engineering: Computer

Artificial Intelligence

ENGINEERING EXPERIENCE

Summer Internship

Itasca, IL

June 2025 – August 2025

Leister Technologies

- Conducted research and executed test runs to evaluate and implement automation solutions using Excel VBA, Javabased macros, and GUI tools in HTML/JavaScript, whilst paired with AI to improve internal software processes.
- Collaborated with engineering teams through tool repair trainings and technical sessions, gaining hands-on exposure to product design, diagnostics, and industrial heating tool applications.
- Represented the company at major trade show SUR/FIN, networking with partner firms and industry leaders to expand professional connections and strengthen technical knowledge of emerging engineering technologies.

Computer Builder & Consultant

Oakbrook, IL

May 2020 – Present

Personal Business

- Assembled and configured 30+ custom-built computers, integrating ideal hardware components to meet diverse performance and budget needs while ensuring exceptional customer satisfaction.
- Conducted 10+ educational workshops, guiding individuals of varying skill levels in building computers from scratch.
- Spearheaded multiple client projects simultaneously which focused on overclocking and optimizing computer hardware, resulting in improved system performance and reliability.

Hand-Voice Controlled Drone System

Iowa City, IA

April 2026 – May 2026

IoT Drone Control & Simulation Project

- Built an IoT-based drone simulation system using Arduino Nano 33 BLE, Raspberry Pi, Webots, and the Crazyflie quadcopter model.
- Developed a gesture-control pipeline that processed IMU roll, pitch, and yaw data with Madgwick filtering and transmitted motion commands over BLE and UDP.
- Integrated microphone-based whistle input for altitude control and logged drone telemetry, including position and orientation data, to InfluxDB for flight-path analysis.

WORK EXPERIENCE

Pool Manager

Oak Brook, IL

August 2019 – Present

Oak Brook Park District Aquatic Center

- Manage a team of lifeguards and oversee all pool operations, including water chemistry, pump room oversight, and facility maintenance, while contributing to management meetings for strategic facility planning.
- Ensure a safe and welcoming environment by addressing patron needs, resolving issues, and upholding high customer service standards, prioritizing safety and customer satisfaction.

IT Contractor

Tinley Park, IL

July 2021 – August 2026

EPS Rehabilitation Inc

- Disassembled 7 malfunctioning computers and reconfigured them into 4 operational systems, optimizing resource utilization.
- Performed maintenance on existing computers, updating software and installing new applications to enhance system performance and reliability.

SKILLS

Java

Python

C/C++

Assembly

Embedded

Breadboarding

MATLAB